SOLVING SYSTEMS OF LINEAR EQUATIONS

LINEAR ALGEBRA

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Linear Equation

System of Linear Equations

Solving System of Linear Equations

- By Substitution
- By Elimination

LINEAR EQUATION

$$\boldsymbol{a_1x_1 + a_2x_2 + \cdots + a_nx_n = b}$$

An equation that can be expressed as the **sum of products** of coefficients and variables, such that the sum equals a **constant**

$x_1, x_2, \ldots, x_n$ are the variables

$a_1, a_2, \ldots, a_n$ are the coefficients

b is the constant

LINEAR EQUATION

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

$$2x - 3y + 4z = 45$$

is a linear equation

x, y, z are the variables

$2, -3, 4$ are the coefficients

45 is the constant

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

square of a variable

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

C. $5a + 2 = 6b - \sqrt{2}c$

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

product of variables

C. $5a + 2 = 6b - \sqrt{2}c$

Which of the following is a linear equation?

A. $4x^2 + 5x + 6 = 0$

B. $\frac{1}{3}uv + \frac{2}{3}u = \frac{3}{5}v$

C. $5a + 2 = 6b - \sqrt{2}c$

$5a - 6b + \sqrt{2}c = -2$

coeff · variable

SYSTEM OF LINEAR EQUATIONS

SYSTEM OF LINEAR EQUATIONS

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

A **set** of linear equations in the **same variables**

SOLUTION

$$2a + 3b + c = 11$$

$$b + 2c = 8$$

$$a + 3c = 10$$

A **set of values** that satisfies every equation in the system

$$a = 1$$

$$b = 2$$

$$c = 3$$

is a solution

SOLVING SYSTEM OF LINEAR EQUATIONS

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

Solve the system of linear equations

$$a - 2b = -10$$

$$3a + 5b = 14$$

Solve the system of linear equations

(1) by **substitution**

(2) by **elimination**

$$2a - 4b - 2c = 12$$

$$-a + 2b - c = 0$$

$$5a - 5b + 3c = 6$$

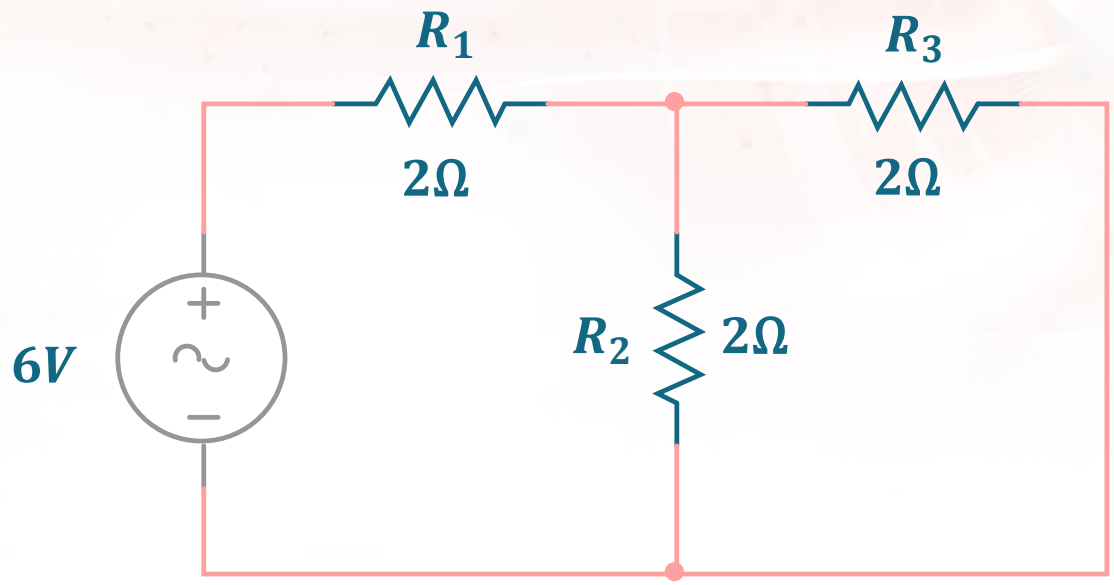
SEATWORK

PROBLEM 1

Determine the voltages v_1, v_2 and v_3 of the given network by

(1) **substitution**

(2) **elimination**



$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$v_1 + v_2 + 0 = 6$$

$$0 - v_2 + v_3 = 0$$